AI-POWERED SENTIMENT ANALYSIS SYSTEM

Month 2 Project – Evaluation Report

1. Abstract

An end-to-end supervised NLP system classifies customer reviews as Positive, Negative or Neutral. The workflow covers data validation, EDA, preprocessing, TF-IDF and BoW, model comparison, grouped cross-validation, tuning, evaluation, serialization and Streamlit deployment.

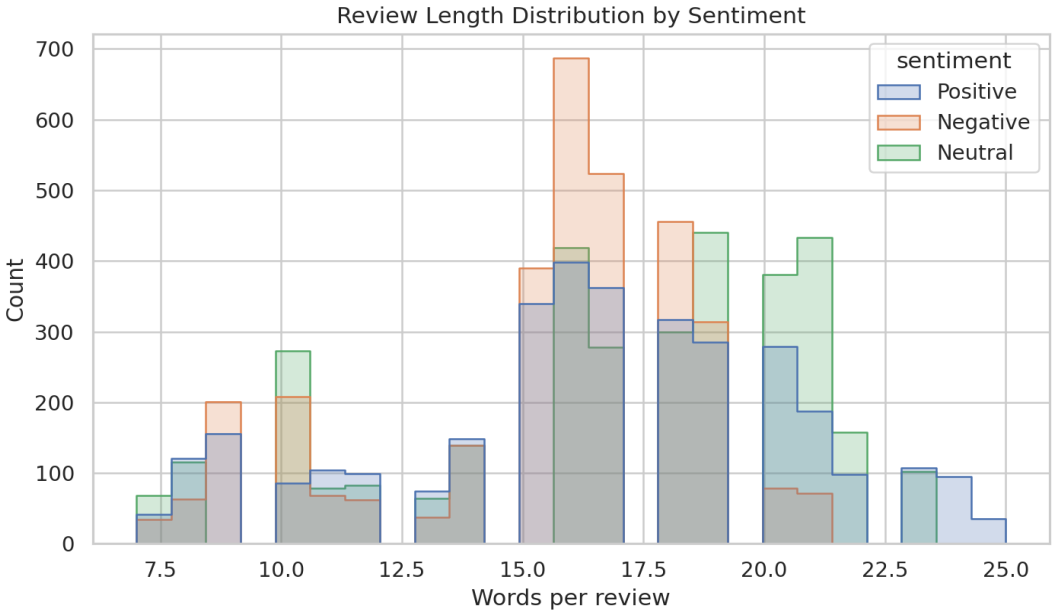
2. Dataset and data quality

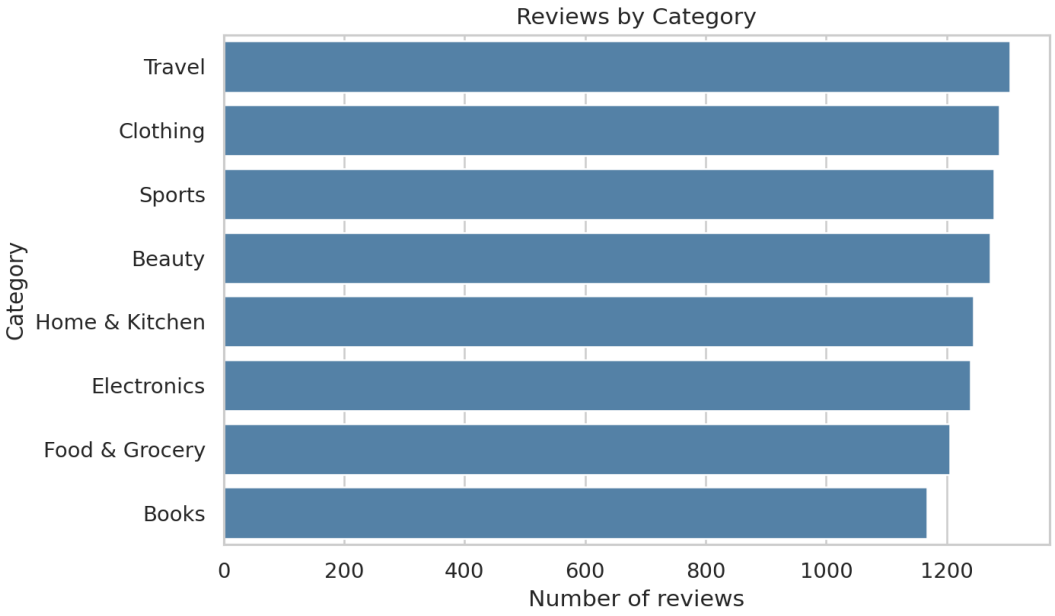
The CSV contains 10,000 rows and five columns. No missing values or exact duplicate rows were found. Sentiment counts: Positive 3,334, Negative 3,333, Neutral 3,333. A key limitation is that there are only 300 distinct review strings across 10,000 rows; 9,700 rows repeat an existing review text. The original rows were retained, and identical texts were grouped during the 80/20 split and five-fold CV to prevent direct text leakage.

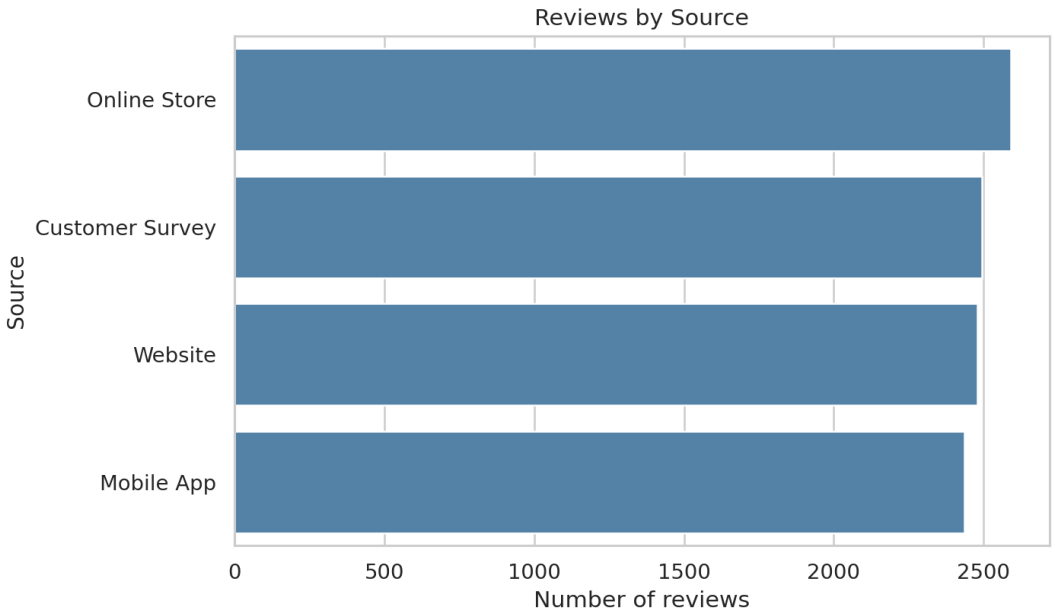
3. Exploratory data analysis

Mean review length: 16.30 words; median: 17 words. The sentiment classes are nearly balanced.




Review Length Distribution by Sentiment
sentiment
Positive
Negative
Neutral
Count
0
100
200
300
400
500
600
700
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0
Words per review


Reviews by Category
Category
Travel
Clothing
Sports
Beauty
Home & Kitchen
Electronics
Food & Grocery
Books
0
200
400
600
800
1000
1200
Number of reviews



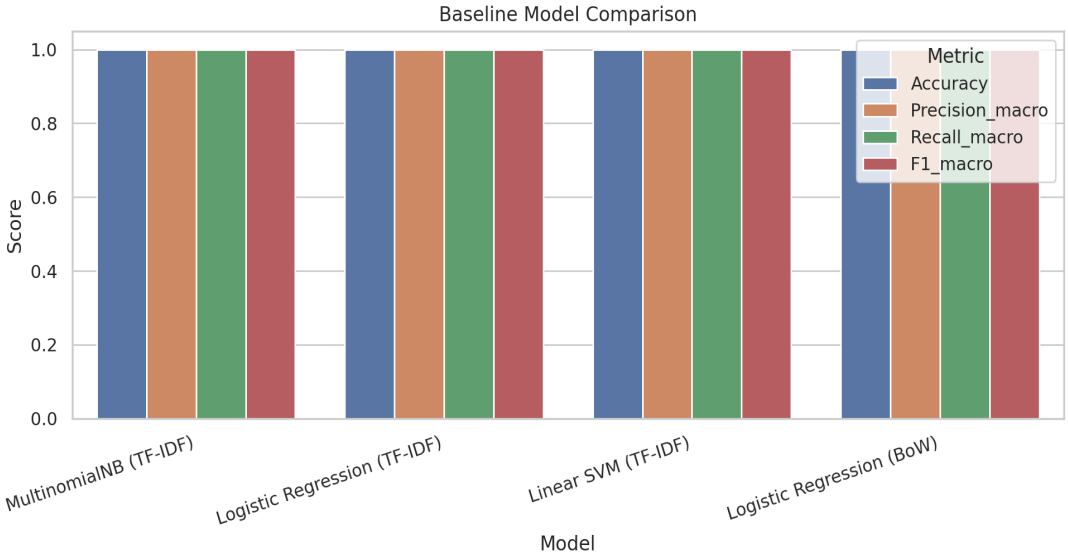
4. Preprocessing and feature extraction

The text normalizer decodes HTML entities, removes tags, replaces URLs, lowercases, removes most punctuation and normalizes whitespace. Negation terms are retained. BoW token counts and TF-IDF term weights were compared. Vectorizers are fitted inside pipelines to avoid vocabulary leakage.

5. Model comparison

Baseline classifiers: Multinomial Naive Bayes + TF-IDF, Logistic Regression + TF-IDF, Linear SVM + TF-IDF, and Logistic Regression + BoW. All achieved the same reported holdout scores on this corpus:

Model	Accuracy	Macro P	Macro R	Macro F1
MultinomialNB (TF-IDF)	1.000	1.000	1.000	1.000
Logistic Regression (TF-IDF)	1.000	1.000	1.000	1.000
Linear SVM (TF-IDF)	1.000	1.000	1.000	1.000
Logistic Regression (BoW)	1.000	1.000	1.000	1.000

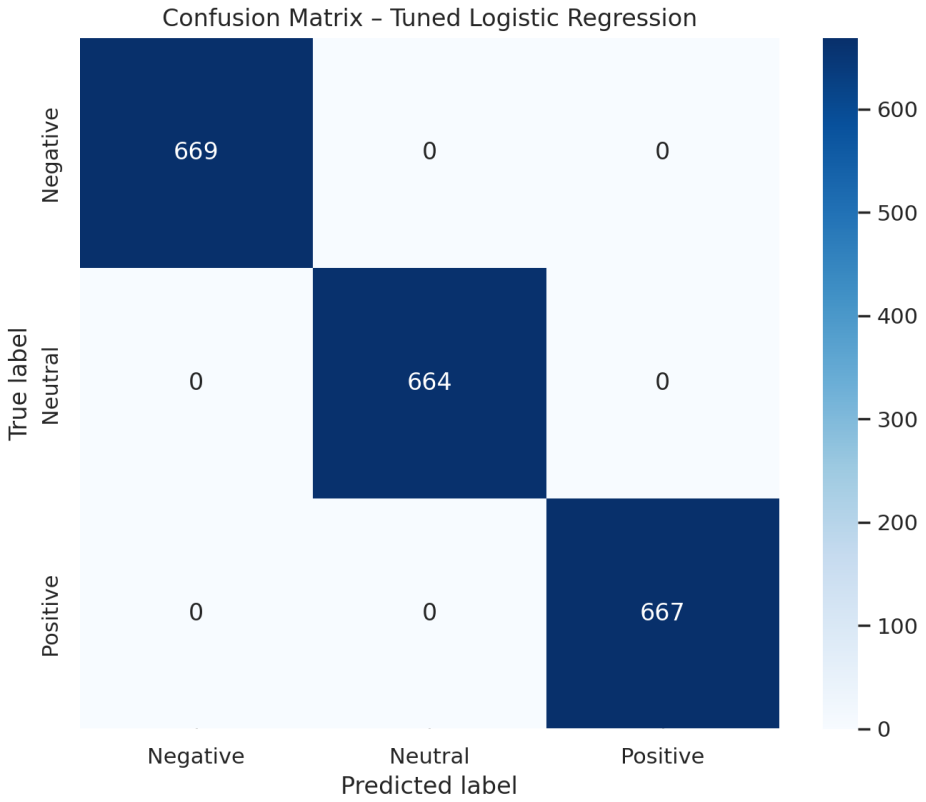


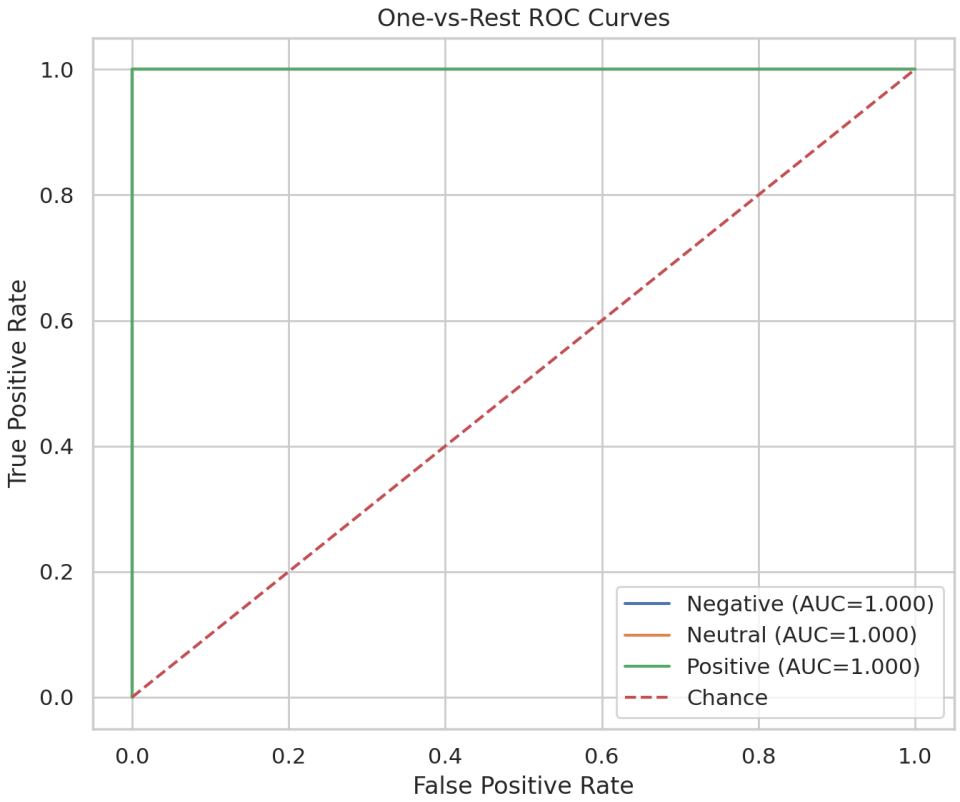
6. Tuning and cross-validation

GridSearchCV used five-fold StratifiedGroupKFold and macro F1. Best parameters: C=0.5, ngram_range=(1, 1), min_df=1. Best CV macro F1=0.9964. Mean CV metrics: accuracy 0.9964, precision_macro 0.9966, recall_macro 0.9964, f1_macro 0.9964.

7. Final evaluation

On 2,000 grouped holdout rows: accuracy 1.0000; macro precision 1.0000; macro recall 1.0000; macro F1 1.0000; macro one-vs-rest ROC-AUC 1.0000. Precision measures correctness among predicted class members; recall measures detection among actual class members; F1 is their harmonic mean.





8. Overfitting / underfitting

The best candidate's training-fold macro F1 was 1.0000, compared with grouped validation macro F1 0.9964; gap=0.0036. The small gap suggests limited overfitting within these folds. Both scores are high, so there is no clear underfitting signal on this corpus. Nevertheless, the repeated-template limitation means this does not establish broad real-world generalization.

9. Deployment

The selected TF-IDF + Logistic Regression pipeline is saved to models/sentiment_pipeline.joblib. app.py loads it and returns a predicted class and estimated class probabilities. Run: pip install -r requirements.txt; then streamlit run app.py.

10. Conclusion and limitations

The project fulfils the core supervised NLP workflow and supplies a reusable Streamlit prototype. The exceptionally high scores must be qualified: only 300 unique review strings occur in the supplied 10,000-row dataset. Real-world use requires a larger, diverse, independently authored and human-labelled corpus plus external validation and probability calibration.